

# EchoFuseNet - Remaining Work

What is done, what is left, and what to do next

AI System for Personalized Cardiac Risk Prediction | Scope: Paper 1 (EchoFuseNet) + Paper 2 (CausalEchoNet) | 2026-07-07

## Bottom line

The framework is ~75% built and fully unit-tested (268 tests pass). Almost all engineering is done. What remains is not mostly code - it is compute runs, two dataset downloads, a few optional modules, and the manuscript. One dependency gates most of it.

### CRITICAL BLOCKER - everything downstream waits on this

The Paper 1 GPU 'headline' training run has not happened yet. Until it does and the trained encoder is frozen and exported, Paper 2 cannot produce any results, and no final tables/figures can be locked. This is the single highest-priority item.

## Remaining work

| # | Task                                                    | What is needed                                                             | Priority   |
|---|---------------------------------------------------------|----------------------------------------------------------------------------|------------|
| 1 | Paper 1 GPU headline run + freeze/export encoder        | 30-epoch oversampled training; export frozen checkpoint. Unblocks Paper 2. | CRITICAL   |
| 2 | 5-fold patient-grouped cross-validation                 | Run on GPU; report mean +/- CI. Code ready.                                | HIGH       |
| 3 | 7-config modality ablation + McNemar table              | Run single / pairwise / full branches. Code ready.                         | HIGH       |
| 4 | INCART external validation run                          | Download INCART DB, run existing pipeline.                                 | HIGH       |
| 5 | PTB-XL domain-transfer run                              | Download PTB-XL, run existing probe (NORM-subset N-recall).                | HIGH       |
| 6 | Paper 2 end-to-end run                                  | Code complete but 0 results - blocked on task #1 frozen encoder.           | BLOCKED    |
| 7 | print() -> logging migration                            | Small repo-wide code-quality cleanup. Non-blocking.                        | LOW        |
| 8 | Manuscript: lock recipe, figures, tables w/ CIs, submit | Results tables + significance; BSPC (P1) / IEEE JBHI (P2).                 | AFTER RUNS |

## Optional - needs supervisor sign-off before building

These are legitimate ideas but sit outside the approved 2-paper scope, or do not yet exist in the repo. Do not build silently:

| Item                            | Note                                                    |
|---------------------------------|---------------------------------------------------------|
| Noise-robustness module         | Does NOT exist yet (see correction below).              |
| ONNX export + quantization      | Does NOT exist yet; only size/latency benchmark exists. |
| Self-supervised ECG pretraining | Changes proposal methodology - flag first.              |
| Test-time adaptation            | Changes what 'the model' is at inference - flag first.  |
| Knowledge distillation          | Implies a larger teacher model - flag as experiment.    |

| Item                                | Note                                               |
|-------------------------------------|----------------------------------------------------|
| Full domain adaptation (DANN/CORAL) | Only a transfer probe exists; bigger claim - flag. |

### Correction worth knowing

REMOVE\_PAPER3\_GUIDE.md claims noise-robustness (robustness/noise.py) and deployment (deployment/onnx\_export.py) are already done. They are NOT in the repo. If those results are wanted, the modules still need to be written and run.

## Already done (do not rebuild)

For reassurance - these checklist items are implemented and unit-tested:

| Area                          | What is in place                                                               |
|-------------------------------|--------------------------------------------------------------------------------|
| Imbalance handling            | Focal loss, class-balanced loss, weighted-CE, balanced sampler.                |
| Full metrics                  | Precision, recall, F1 (macro + per-class), MCC, Cohen's kappa.                 |
| Hyperparameter opt            | Optuna TPE search (leakage-safe DS1-internal CV objective).                    |
| Uncertainty                   | MC Dropout + temperature scaling + ECE. (Deep ensemble deliberately excluded.) |
| Statistical tests             | Wilcoxon, McNemar, bootstrap CIs, paired t-test.                               |
| Baseline table                | ResNet18/50, DenseNet121, EfficientNet-B0, ConvNeXt, ViT-B/16.                 |
| Training engineering          | AMP, early stopping, cosine LR, EMA, grad clipping, checkpoint averaging.      |
| Reproducibility               | Seeds everywhere; per-run experiment log (git commit, config, seed, metrics).  |
| Explainability (Paper 2)      | Grad-CAM, exact SHAP, per-modality causal ITE attribution.                     |
| External-validation framework | INCAART + PTB-XL loaders and scripts (built; awaiting data + run).             |

## Is Paper 2 complete?

**Code: yes. Results: no - and it is blocked.**

All Paper 2 modules are present and tested (7 test files green): frozen-encoder loader, feature-space CVAE + counterfactual metrics, per-modality ITE attribution, Grad-CAM/SHAP baselines, representation caching, and a wired end-to-end training loop. But the training entry point is runtime-gated: it refuses to run unless it is given a FROZEN Paper 1 checkpoint. That checkpoint does not exist because task #1 (the GPU headline run) has not happened. So Paper 2 = 100% framework, 0% delivered results, until Paper 1's encoder is trained and frozen.

## Recommended order

1. Run the Paper 1 GPU headline training; freeze and export the encoder. (Unblocks everything.)
2. In parallel: download INCAART + PTB-XL and run the external-validation pipelines.
3. Run 5-fold CV and the modality ablation; build results tables with CIs + significance.
4. Run Paper 2 end-to-end against the frozen encoder; generate counterfactual/attribution results.
5. Do the print -> logging cleanup (small, non-blocking).
6. Write manuscripts; decide with supervisor on any optional modules (noise/quantization/SSL...).